

Reachy Pill Count Comparison Test

Mentor-ready summary cleaned from the raw source tab | May 27, 2026

Overall count-match accuracy	14 of 20 trials	Using the corrected Easy ground-truth count of 5									
Remaining mismatches	6 of 20 trials	All remaining misses are in Hard or Hard - Closer layouts									
Most reliable setups	Easy; Medium; Hard - No Overlap	All 4 conditions matched after the Easy correction									
Main failure mode	Dense / closer-packed trays	Hard C/D and all Hard - Closer rows still undercounted									
Trial	Setup	Condition	Image Input	Run Path	Ground Truth Count	Returned Count	Diff	Status	Ground Truth Items	Returned Items	Notes
1	Easy	A	Phone/Laptop Camera	Gemini App	5	5	0	Match	2 brown, 2 white, 1 green	2 white; 2 tan/brown; 1 green	
2	Easy	B	Phone/Laptop Camera	Reachy Code (test_vlm.py)	5	5	0	Match	2 brown, 2 white, 1 green	2 white; 2 brown; 1 green	
3	Easy	C	Reachy Snapshots (3)	Gemini App	5	5	0	Match	2 brown, 2 white, 1 green	2 white; 2 brown; 1 dark green	
4	Easy	D	Reachy Camera	Reachy Full Scan	5	5	0	Match	2 brown, 2 white, 1 green	2 white; 2 tan/brown; 1 dark green	
5	Medium	A	Phone/Laptop Camera	Gemini App	7	7	0	Match	3 brown, 2 white, 2 green	3 tan/brown; 2 white; 2 dark green	
6	Medium	B	Phone/Laptop Camera	Reachy Code (test_vlm.py)	7	7	0	Match	3 brown, 2 white, 2 green	3 tan/brown; 2 white; 2 green	
7	Medium	C	Reachy Snapshots (3)	Gemini App	7	7	0	Match	3 brown, 2 white, 2 green	3 tan/brown; 2 white; 2 green	
8	Medium	D	Reachy Camera	Reachy Full Scan	7	7	0	Match	3 brown, 2 white, 2 green	3 tan/brown; 2 white; 2 green	
9	Hard	A	Phone/Laptop Camera	Gemini App	10	10	0	Match	4 brown, 3 white, 3 green	4 light brown; 3 white; 3 dark green	
10	Hard	B	Phone/Laptop Camera	Reachy Code (test_vlm.py)	10	10	0	Match	4 brown, 3 white, 3 green	4 brown; 3 green; 3 white	
11	Hard	C	Reachy Snapshots (3)	Gemini App	10	9	-1	Mismatch	4 brown, 3 white, 3 green	3 tan/brown; 3 white; 3 black	Undercount by 1 in the Reachy snapshot route.
12	Hard	D	Reachy Camera	Reachy Full Scan	10	9	-1	Mismatch	4 brown, 3 white, 3 green	3 brown; 3 white; 3 black	Undercount by 1 in the full-scan route.
13	Hard - Closer	A	Phone/Laptop Camera	Gemini App	10	9	-1	Mismatch	Not listed in source row	3 tan/brown; 3 white; 3 dark green	Source row was shifted left; standardized here.
14	Hard - Closer	B	Phone/Laptop Camera	Reachy Code (test_vlm.py)	10	9	-1	Mismatch	Not listed in source row	3 tan/brown; 3 white; 3 green	Source row was shifted left; standardized here.
15	Hard - Closer	C	Reachy Snapshots (3)	Gemini App	10	9	-1	Mismatch	Not listed in source row	3 tan/brown; 3 white; 3 dark green	Source row was shifted left; standardized here.
16	Hard - Closer	D	Reachy Camera	Reachy Full Scan	10	8	-2	Mismatch	Not listed in source row	3 brown; 3 dark green; 2 white	Largest miss in the sheet: full scan returned 8 vs 10.
17	Hard - No Overlap	A	Phone/Laptop Camera	Gemini App	10	10	0	Match	Not listed in source row	4 tan/brown; 3 white; 3 dark green	Source row was shifted left; standardized here.
18	Hard - No Overlap	B	Phone/Laptop Camera	Reachy Code (test_vlm.py)	10	10	0	Match	Not listed in source row	4 tan/brown; 3 white; 3 dark green	Source row was shifted left; standardized here.
19	Hard - No Overlap	C	Reachy Snapshots (3)	Gemini App	10	10	0	Match	Not listed in source row	4 brown; 3 white; 3 green	Source row was shifted left; standardized here.
20	Hard - No Overlap	D	Reachy Camera	Reachy Full Scan	10	10	0	Match	Not listed in source row	4 brown; 3 white; 3 green	Source row was shifted left; standardized here.

What these results show

- Medium and Hard - No Overlap were stable across all four conditions, which suggests the prompt/model can work when pills are visually separable.

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- The biggest reliability drop happens in dense or closer-packed layouts: Hard C/D undercounted by 1, and every Hard - Closer condition undercounted, with full scan dropping to 8 vs 10.
- Next process change: verify ground-truth labels first, then focus iteration on capture quality and clutter-handling prompts rather than broad prompt changes across every scenario.

[illegible]